

BANARAS HINDU UNIVERSITY

B.A./B.Sc. (Hons.) Semester V Examination, 2022-23

Paper No. STA/STB-501: Applied Statistics

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer any **five** questions. All questions carry equal marks. Symbols have their usual meanings.*

Question 1.

- (a) What are the methods of obtaining Vital Statistics? Also give the uses of Vital Statistics. [7]
- (b) Define Gross Reproduction Rate (GRR) and Net Reproduction Rate (NRR). Explain how far they can be considered as indices of population growth. Interpret the following cases:
- (i) $GRR = 2.10$, $NRR = 1.10$
 - (ii) $GRR = 1.15$, $NRR = 1.10$
 - (iii) $GRR = 1.75$, $NRR = 1.80$

[7]

Question 2.

- (a) Discuss about the logistic curve used for measuring the growth of a population. [7]
- (b) What do you mean by a life table? Write the assumptions of a life table clearly. Discuss the various columns of a complete life table and also give the uses of a life table. [7]

Question 3.

- (a) Define crude death rate and point out its limitations. Explain clearly the standardised death rates, including the methods of their computation. [7]
- (b) Describe the different measures of fertility with their merits and demerits. [7]

Question 4.

- (a) Discuss the importance and limitations of index numbers. [7]
- (b) What is the need for a family budget enquiry in the construction of the cost of living index number? [7]

Question 5.

- (a) Define the time reversal test. Show that Fisher's index number satisfies the time reversal test. [7]
 - (b) What is statistical quality control (SQC)? Describe the main principles of SQC charts. Discuss the need and utility of SQC in industry. [7]
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Question 6.

- (a) Describe control charts for mean ($\bar{X}$ -chart) and standard deviation (S -chart). [7]
 - (b) Discuss in detail the construction of p -chart and c -chart for attributes. [7]
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Question 7.

- (a) Define infant mortality rate. Why do we need to adjust the infant mortality rate? Describe in detail the Logan method of adjustment of the infant mortality rate. [7]
 - (b) Explain secular trend in detail. Critically examine the various methods of measuring trend. [7]
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Question 8.

- (a) Write short notes on the different types of components of time series and explain the difference between seasonal and cyclical variations with examples. [7]
 - (b) What do you mean by time series? Describe its uses in detail. [7]
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Question 9.

- (a) Explain the additive and multiplicative models of time series, stating clearly the assumptions underlying each model. [7]
 - (b) Explain the moving average method of trend measurement. How does one choose the appropriate length of the moving average? [7]
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